

# 1 Executive Summary

## The Business in Three Sentences

**NegoPlay** is a B2B SaaS platform that automatically discovers decision-making profiles from elite competitive data and deploys them as AI agents to help professionals prepare for, simulate, and win business negotiations.

The core insight: 149,208 expert bridge hands reveal stable behavioral archetypes (“Slam Hunter”, “Insurance Player”, etc.) that transfer measurably to negotiation behavior. We turn this empirical finding into a subscription API for sales teams, procurement departments, and M&A advisors.

**Target revenue in Year 3:** \$480K ARR. **Total budget for MVP:** under \$15 (LLM costs only, excluding developer time).

Table 1.1: Business plan summary.

| Category         | Value                                                |
|------------------|------------------------------------------------------|
| Market (TAM)     | \$4.2B (negotiation intelligence + sales AI)         |
| Market (SAM)     | \$420M (mid-market + enterprise, English-speaking)   |
| Market (SOM Y3)  | \$480K ARR (realistic 5-year ramp)                   |
| Business model   | B2B SaaS subscription + usage-based API              |
| Primary customer | Sales directors, procurement managers, M&A teams     |
| Pricing (entry)  | \$49 / month per seat                                |
| Competitive moat | Empirically validated profiles from real expert data |
| Technology stack | Python, Gemini Flash 2.0, scikit-learn, GitHub       |

## 2 The Problem

### 2.1 Negotiations fail because we do not know our counterpart

Every business negotiation involves incomplete information. We know our own BATNA (Best Alternative to a Negotiated Agreement) but rarely know the other side's risk tolerance, decision style, or likely concession pattern. The result:

- Deals are left on the table when one side concedes too quickly.
- Deals collapse when one side misjudges the other's walk-away point.
- Sales cycles are prolonged because reps cannot adapt their style to the buyer's decision profile.

### 2.2 The data gap

Business negotiation lacks the clean feedback loops that make other domains trainable:

- Real negotiation transcripts are proprietary.
- Outcomes ("who won?") are rarely measurable in standardized form.
- No public dataset allows training behavioral models on real expert negotiation decisions.

Existing tools (Gong, Chorus, Clari) analyze *what was said* after the fact. No tool predicts *how the counterpart will behave* before the conversation starts.

#### The gap we fill

NegoPlay is the first platform to build behavioral profiles from a large, public, measurable expert decision dataset – bridge tournament records – and transfer those profiles to negotiation preparation.

## 3 The Solution

### 3.1 What NegoPlay does

NegoPlay discovers four to five decision-making archetypes from 149,208 elite bridge tournament hands. Each archetype is translated into a named profile with five to seven behavioral traits. The profiles are then deployed as AI agents that can:

1. **Profile your counterpart** – given publicly available information about the person or company, classify them into the nearest archetype.
2. **Simulate the negotiation** – run a multi-round AI-vs-AI simulation of the upcoming deal so the user can see likely concession patterns before the real meeting.
3. **Advise in real time** – via API integration with CRM tools (HubSpot, Salesforce), push profile-based suggestions during a live call.

## 3.2 The four profiles (confirmed by Stage 1)

Table 3.1: Decision profiles discovered from 149,208 bridge tournament hands. Profile names and assignment rates are the actual Stage 1 outputs (v2.1, May 2026).

| Profile               | $n$ (%)      | Core traits (from bridge data)                                                                                 | Predicted negotiation behavior                                                                 |
|-----------------------|--------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Slam Hunter           | 20 (3.6 %)   | Bids slams at $1.84\times$ the population baseline; calculated upside bets; trusts partner’s strength signals. | Opens high, concedes little, walks away fast if the value gap is large.                        |
| Insurance Player      | 21 (3.7 %)   | Stays in partscore at $1.20\times$ baseline; risk-averse; values certainty over upside.                        | Prefers small guaranteed gains; long close cycles; many contingencies in the contract.         |
| NT Specialist         | 17 (3.0 %)   | Plays NoTrump contracts at $1.36\times$ baseline; analytical; precise trick counting.                          | Data-driven; demands DCF / comparables justification; rejects “gut-feel” offers.               |
| Fighter               | 37 (6.6 %)   | Doubles opponents for penalties at $1.55\times$ baseline; competitive; aggressive against opponents.           | Counter-offers aggressively; threatens walk-away as a tactic; rejects first offer reflexively. |
| Generalist (baseline) | 468 (83.1 %) | Average rates on all axes; control population.                                                                 | Default-reasonable; matches counterpart’s style.                                               |

*Note on naming consistency.* Earlier internal drafts of this plan used the working names “Bluffer” and “Doubler” for two of the profiles. After Stage 1 ran on the full 149K-row dataset, the actual statistical axes turned out to be `nt_rate` and `penalty_double_rate`, leading to the more descriptive names *NT Specialist* and *Fighter*. The Preliminary Report (Table A.2) and this Business Plan are now synchronised on these names.

## 3.3 Why bridge data works

Bridge is one of the only public, large-scale, measurable records of expert decision-making under incomplete information with four-player dynamics (two cooperating, two competing – exactly like a business negotiation involving two companies each with internal stakeholders). The IMP scoring system provides a clean, standardized outcome signal. No comparable dataset exists for business negotiation.

## 4 Market Analysis

### 4.1 Total Addressable Market (TAM)

The global sales intelligence software market was valued at approximately \$3.4B in 2023 and is growing at 11 % CAGR. Adjacent markets include procurement analytics (\$1.2B) and M&A advisory tools (\$0.6B). We define NegoPlay’s TAM as the intersection of these segments:

Table 4.1: TAM breakdown.

| Segment                         | Market size (2024) | CAGR        |
|---------------------------------|--------------------|-------------|
| Sales intelligence (enterprise) | \$2.8B             | 11 %        |
| Procurement AI tools            | \$0.9B             | 15 %        |
| M&A deal intelligence           | \$0.5B             | 9 %         |
| <b>Total TAM</b>                | <b>\$4.2B</b>      | <b>11 %</b> |

### 4.2 Serviceable Addressable Market (SAM)

NegoPlay targets mid-market and enterprise companies in English-speaking markets (US, UK, Australia, Canada) that: (a) use a CRM, (b) have active outbound sales or procurement teams, and (c) have an existing AI software budget. This narrows the addressable market to approximately 10 % of the TAM.

Table 4.2: SAM calculation.

| Parameter                      | Value           |
|--------------------------------|-----------------|
| TAM                            | \$4.2B          |
| English-speaking market share  | × 40 %          |
| Mid-market + enterprise filter | × 60 %          |
| AI-ready companies             | × 45 %          |
| <b>SAM</b>                     | <b>≈ \$450M</b> |

### 4.3 Serviceable Obtainable Market (SOM)

In the first three years, NegoPlay targets 0.1 % of the SAM through direct sales to early adopters (academic networks, bridge-community connections, AI-forward SMEs).

Table 4.3: SOM projection over three years.

| <b>Metric</b>                  | <b>Year 1</b> | <b>Year 2</b> | <b>Year 3</b> |
|--------------------------------|---------------|---------------|---------------|
| Paying customers (seats)       | 20            | 120           | 800           |
| Average revenue per seat/month | \$49          | \$55          | \$50          |
| Monthly Recurring Revenue      | \$980         | \$6,600       | \$40,000      |
| Annual Recurring Revenue (ARR) | \$11,760      | \$79,200      | \$480,000     |

# 5 Business Model and Pricing

## 5.1 Revenue model

NegoPlay uses a **freemium-to-subscription** model with an optional usage-based API tier for enterprise integrations.

Table 5.1: Pricing tiers.

| Tier       | Audience                           | Includes                                             | Price                                |
|------------|------------------------------------|------------------------------------------------------|--------------------------------------|
| Free       | Individual / researcher            | 10 profile analyses/month, no simulation             | \$0                                  |
| Pro        | Sales rep / procurement officer    | 200 analyses + unlimited simulations + email support | \$49/seat/month                      |
| Team       | SME sales team (up to 20 seats)    | Unlimited analyses, CRM web-hook, team dashboard     | \$199/month flat                     |
| Enterprise | Large enterprise / API integration | Unlimited, SLA, SSO, Salesforce / HubSpot connector  | Custom (est. \$2,000–\$10,000/month) |

## 5.2 LLM pricing model – input vs. output tokens

LLM providers price input tokens (the prompt the model reads) and output tokens (the response the model generates) at *different* rates. Output tokens are typically 4× more expensive because generation requires more compute per token than reading. Any honest cost model must therefore

track both directions separately.

Table 5.2: Per-million-token pricing across LLM providers (May 2026). Output is consistently more expensive than input.

| Provider / model                  | Input       | Output      | Out/In ratio |
|-----------------------------------|-------------|-------------|--------------|
| <b>Gemini Flash 2.0 (default)</b> | \$0.075 / M | \$0.30 / M  | <b>4.0×</b>  |
| Gemini 2.5 Pro                    | \$1.25 / M  | \$5.00 / M  | 4.0×         |
| Claude Haiku 4.5                  | \$1.00 / M  | \$5.00 / M  | 5.0×         |
| Claude Opus 4.7                   | \$15.00 / M | \$75.00 / M | 5.0×         |
| GPT-4o                            | \$2.50 / M  | \$10.00 / M | 4.0×         |
| GPT-4o mini                       | \$0.15 / M  | \$0.60 / M  | 4.0×         |

The cost of a single LLM call therefore decomposes as

$$C_{\text{call}} = \frac{T_{\text{in}}}{10^6} \cdot c_{\text{in}} + \frac{T_{\text{out}}}{10^6} \cdot c_{\text{out}} \quad (5.1)$$

where  $T_{\text{in}}, T_{\text{out}}$  are the input and output token counts and  $c_{\text{in}}, c_{\text{out}}$  are the per-million-token unit prices.

### 5.3 Unit economics per analysis

The core cost driver is LLM token consumption. Each NegoPlay profile analysis sends a long prompt (system instruction + counterpart context) and receives a relatively short structured response. The input/output breakdown matters because input dominates the token *count* but output dominates the token *cost*.

Table 5.3: Unit economics per profile analysis, with input and output costs separated. Note that 38 % of total tokens are output, but they account for 71 % of the LLM cost.

| Component                                          | Tokens | Unit rate   | Cost                              |
|----------------------------------------------------|--------|-------------|-----------------------------------|
| Gemini Flash 2.0 – <b>input</b> (system + user)    | 3,000  | \$0.075 / M | \$0.000225                        |
| Gemini Flash 2.0 – <b>output</b> (structured JSON) | 1,800  | \$0.300 / M | \$0.000540                        |
| <b>LLM subtotal per analysis</b>                   | 4,800  | —           | <b>\$0.000765</b>                 |
| Infrastructure (hosting, storage, logging)         | —      | —           | \$0.005                           |
| <b>Total cost per analysis</b>                     | —      | —           | <b>\$0.0058</b>                   |
| <b>Revenue per analysis (Pro)</b>                  | —      | —           | <b>\$0.245</b> (200 calls / \$49) |
| <b>Gross margin per analysis</b>                   | —      | —           | <b>97.6 %</b>                     |



## 5.4 Token economics summary across pipeline stages

Table 5.4: Token economics across the four pipeline stages, with input and output costs separated. The Stage 2 skill-extraction call is the most expensive single call because the LLM reads many bridge hands (high  $T_{in}$ ); per-turn negotiation calls are dominated by output cost.

| Stage                                    | $T_{in}$      | $T_{out}$    | In cost           | Out cost          | Total             |
|------------------------------------------|---------------|--------------|-------------------|-------------------|-------------------|
| Stage 2 – Skill extraction               | 7,000         | 1,000        | \$0.000525        | \$0.000300        | \$0.000825        |
| Stage 3 – Agent prompt assembly          | 1,600         | 400          | \$0.000120        | \$0.000120        | \$0.000240        |
| Stage 4 – Bridge bidding turn            | 2,500         | 500          | \$0.000188        | \$0.000150        | \$0.000338        |
| Stage 4 – Negotiation turn               | 3,500         | 1,500        | \$0.000263        | \$0.000450        | \$0.000713        |
| <b>Full pipeline (one user analysis)</b> | <b>14,600</b> | <b>3,400</b> | <b>\$0.001096</b> | <b>\$0.001020</b> | <b>\$0.002116</b> |

### Why this split matters commercially

**18 % of NegoPlay’s tokens are output, but they account for 48 % of the LLM cost.** Output prompt design is therefore a direct lever on gross margin – returning concise structured JSON instead of verbose free text cuts the per-call cost roughly in half. Our agent prompts (see Listing in the Preliminary Report) require strict JSON schemas of at most 1,500 output tokens for this reason.

### Cost advantage of Gemini Flash 2.0

With **input at \$0.075 / M** and **output at \$0.30 / M**, Gemini Flash 2.0 is approximately 10× cheaper than GPT-4o (\$2.50 / \$10.00) for equivalent throughput, and roughly 200× cheaper than Claude Opus 4.7 (\$15 / \$75). The entire NegoPlay MVP (development + testing) runs for under \$15 in LLM costs. Claude and GPT are used selectively for cross-model validation runs to strengthen publishability, not for production traffic.

# 6 Competitive Landscape

## 6.1 Direct and indirect competitors

Table 6.1: Competitive landscape.

| Competitor           | What they do                                                            | Gap vs NegoPlay                                                                 | Threat     |
|----------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------|------------|
| <b>Gong.io</b>       | Analyzes past sales calls (voice + transcript) to coach reps.           | Post-hoc only; no pre-meeting counterpart profiling; no game-theory basis.      | Medium     |
| <b>Pactum AI</b>     | Autonomous negotiation bot for procurement (IKEA, Walmart).             | Fully automated (no human in the loop); no behavioral profiling of individuals. | Low–Medium |
| <b>Chorus.ai</b>     | Records and transcribes sales calls; CI insights for managers.          | Same as Gong; reactive, not predictive.                                         | Low        |
| <b>Clari</b>         | Revenue forecasting + deal risk scoring.                                | Forecasting only; no behavioral simulation.                                     | Low        |
| <b>Crystal Knows</b> | DISC personality profiling from LinkedIn data.                          | Not empirically validated on expert decision data; no simulation.               | Medium     |
| <b>NegoPlay</b>      | Profiles + simulates counterpart from empirically validated archetypes. | <i>Only platform with an empirical game-theoretic foundation.</i>               | –          |

## 6.2 Our competitive moat

1. **Empirical foundation:** Profiles are derived from 149,208 expert decisions with measurable outcomes – not from personality questionnaires or self-reported surveys. This is defensible in peer-reviewed literature.
2. **Cross-domain validation:** The alignment finding (Spearman  $\rho \geq 0.7$  between bridge and negotiation behavior) can be published – creating an academic moat.
3. **Provider-neutral architecture:** The unified `llm_client.py` wrapper means NegoPlay is not locked to any single LLM provider.
4. **Data accumulation:** Every new customer interaction refines the profiles – a classic data flywheel.

# 7 Go-to-Market Strategy

## 7.1 Phases

### Phase 1 – Academic validation (Months 1–6, free)

- Publish the alignment finding as a conference paper (AAMAS or IJCAI).
- Open-source the profile discovery code on GitHub to build credibility.
- Target bridge clubs, negotiation researchers, and MBA programs as early adopters and free validators.
- **Goal:** 100 registered free users, 1 academic citation.

### Phase 2 – Paid early access (Months 7–18)

- Launch Pro tier (\$49/seat/month) via Product Hunt and LinkedIn.
- Leverage the author’s networks at Sami Shamoon College and LUT University.
- Offer a 3-month pilot to 3–5 mid-market sales teams at cost.
- **Goal:** 20 paying customers, \$11,760 ARR.

### Phase 3 – Enterprise and API (Months 19–36)

- Build HubSpot and Salesforce connectors.
- Pursue enterprise deals with procurement departments (indirect sales via consultancies).
- Explore white-labeling for negotiation training companies.
- **Goal:** 800 seats, \$480K ARR.

## 7.2 Customer acquisition cost estimate

Given that Phase 1 is entirely inbound (open source + academic press), early customer acquisition cost (CAC) is near zero. In Phase 2, assuming \$2,000/month in LinkedIn advertising and content marketing:

- **CAC (Phase 2):**  $\$2,000 / 20 \text{ customers} = \$100 \text{ per customer}$ .

- **LTV (Pro, 12-month retention):**  $\$49 \times 12 = \$588$ .
- **LTV/CAC ratio:**  $5.9 \rightarrow$  healthy (target  $> 3$ ).

## 8 Roadmap and Milestones

Table 8.1: 18-month product and business roadmap.

| Month | Milestone                                                            | Type       |
|-------|----------------------------------------------------------------------|------------|
| 1–5   | Complete NegoPlay MVP (4-stage pipeline, 4 agents, alignment result) | Technical  |
| 5     | Submit alignment paper to AAMAS / AAAI workshop                      | Academic   |
| 6     | Open-source profile-discovery code on GitHub                         | Marketing  |
| 7     | Launch Pro tier on Product Hunt                                      | Commercial |
| 9     | 20 paying customers; first case study published                      | Commercial |
| 12    | HubSpot webhook integration (beta)                                   | Technical  |
| 15    | First enterprise pilot (procurement team)                            | Commercial |
| 18    | \$79K ARR; Salesforce connector GA                                   | Commercial |
| 24    | Series A readiness assessment or academic spin-off decision          | Strategic  |

# 9 Financial Projections

## 9.1 Three-year revenue projection

Table 9.1: Three-year revenue, cost, and margin projection.

| Item                                         | Year 1   | Year 2   | Year 3    |
|----------------------------------------------|----------|----------|-----------|
| <i>Revenue</i>                               |          |          |           |
| Pro seats (avg)                              | 20       | 120      | 600       |
| Pro revenue                                  | \$11,760 | \$79,200 | \$352,800 |
| Enterprise / API revenue                     | —        | —        | \$127,200 |
| <b>Total ARR</b>                             | \$11,760 | \$79,200 | \$480,000 |
| <i>Costs</i>                                 |          |          |           |
| LLM API – input tokens (\$0.075/M)           | \$182    | \$1,221  | \$7,403   |
| LLM API – output tokens (\$0.300/M)          | \$171    | \$1,155  | \$6,997   |
| LLM API subtotal (Gemini, $\approx 97\%$ GM) | \$353    | \$2,376  | \$14,400  |
| Infrastructure (cloud hosting)               | \$1,200  | \$3,600  | \$12,000  |
| Marketing (LinkedIn, content)                | \$0      | \$24,000 | \$60,000  |
| <b>Total COGS + OpEx</b>                     | \$1,553  | \$29,976 | \$86,400  |
| <b>Gross Profit</b>                          | \$10,207 | \$49,224 | \$393,600 |
| <b>Gross Margin</b>                          | 87 %     | 62 %     | 82 %      |

## 9.2 Funding requirement

Phase 1 requires **zero external capital** (developer is the founder; LLM costs are under \$15 for the full MVP). Phase 2 marketing costs (\$24K/year) can be bootstrapped from Phase 1 revenue. A seed round (\$200K–\$500K) would be sought in Year 2 to fund enterprise sales and engineering headcount.

# 10 Business Risks

Table 10.1: Commercial risk register.

| #   | Risk                                                                    | Lik. | Imp. | Mitigation                                                                                                        |
|-----|-------------------------------------------------------------------------|------|------|-------------------------------------------------------------------------------------------------------------------|
| BR1 | Research finding not supported ( $\rho < 0.5$ ).                        | Med  | High | Null result is publishable; pivot to “bridge coaching AI” (profile analytics without negotiation transfer claim). |
| BR2 | Gong or Salesforce launches a competing feature.                        | Med  | High | Academic moat (peer-reviewed publication) + niche positioning (empirical game-theory basis) differentiates us.    |
| BR3 | LLM providers raise prices significantly.                               | Low  | Med  | Provider-neutral <code>llm_client.py</code> lets us switch to the cheapest provider per task within days.         |
| BR4 | Data privacy regulations block use of player name data.                 | Low  | Med  | Player data is public (tournament websites); we profile <i>negotiation counterparts</i> , not users.              |
| BR5 | The bridge-to-negotiation transfer claim is rejected by peer reviewers. | Med  | Med  | Frame as “proof of concept in simulated environment” from the start; null result is reported honestly.            |
| BR6 | Slow enterprise sales cycle (>9 months).                                | High | Med  | Focus on SME Pro tier first; enterprise is a Year 2+ play.                                                        |

# 11 Team

## Founder Profile: Anna Ben-Shushan

- **Current role:** Lecturer at Sami Shamoon College of Engineering; PhD candidate at LUT University (Finland).
- **Academic:** M.Sc. Software Engineering; AI Development Expert course (current).
- **Industry:** Full-stack developer; AI consultant.
- **PhD topic:** “From Bridge Tables to Global Markets: AI and Optimization Models of Cooperation, Competition, and Negotiation on Digital Platforms.”
- **Unique advantage:** Domain expert in both the data source (bridge) and the application domain (business negotiation), with the academic credibility to publish the alignment finding.

## 11.1 Hiring plan (if funded)

- **Month 9:** Part-time front-end developer (student, paid project).
- **Month 15:** Full-time ML engineer (seed-funded).
- **Month 18:** Business development manager for enterprise sales.



## 12 Conclusion

NegoPlay sits at the intersection of three trends: the rapid adoption of AI in enterprise software, the growing demand for data-driven negotiation tools, and the academic maturation of multi-agent LLM systems.

Its core differentiator is an **empirically grounded behavioral model** – profiles discovered from 149,208 expert decisions rather than assembled from questionnaires. This gives NegoPlay both a commercial moat and an academic story that can be defended in peer review.

The financial model is attractive: 97% gross margins on LLM calls, near-zero CAC in Phase 1, and a clear path to \$480K ARR by Year 3 without external capital for the first 18 months.

### The single most important number

**Spearman  $\rho \geq 0.7$**

If the cross-domain alignment exceeds this threshold, NegoPlay has a publishable result, a defensible product claim, and a credible reason for enterprise customers to pay.

Everything else follows from this number.